# BIOS vs UEFI Comparison

| Category | BIOS | UEFI |
|---|---|---|
| Definition | Traditional firmware that initializes hardware and starts the operating system boot process. | Modern firmware interface designed to replace BIOS with a more flexible and feature-rich boot environment. |
| Boot Process | Uses the legacy boot process and commonly loads boot code from the Master Boot Record (MBR). | Uses a modern boot process and loads boot files from the EFI System Partition (ESP). |
| Maximum Disk Support | Commonly limited to about 2 TB when paired with MBR. | Supports very large disks when paired with GPT. |
| Partition Style | Primarily associated with MBR. | Primarily associated with GPT. |
| Security Features | Limited built-in security compared with modern firmware. | Supports features such as Secure Boot for verifying trusted boot components. |
| Speed | Simple legacy startup process. | Generally faster and more flexible on modern systems. |
| Advantages | Simple, familiar, and compatible with older hardware and operating systems. | Supports GPT, large disks, Secure Boot, modern hardware, and flexible boot management. |
| Disadvantages | Older design, MBR limitations, and fewer modern security capabilities. | May have compatibility issues with very old legacy systems. |
| Modern Usage | Mostly found on older computers and legacy deployments. | Standard firmware interface on most modern computers. |

## Why UEFI Has Largely Replaced BIOS

UEFI has largely replaced BIOS because it addresses several limitations of traditional BIOS firmware. The most important improvement is its support for GPT partitioning, which allows modern computers to use storage devices far larger than the traditional MBR limit and provides a more flexible partition layout.

UEFI also improves platform security. Secure Boot can verify trusted boot components before the operating system starts, helping reduce the risk of unauthorized or malicious software loading during the early boot process. UEFI also provides a more capable boot environment, including boot managers and firmware applications stored on the EFI System Partition.

Modern computers benefit from UEFI because they commonly use large storage devices, newer processors, and operating systems designed around GPT and modern firmware capabilities. BIOS remains useful for legacy compatibility, but UEFI is the preferred firmware standard for current systems.